Empirical Proof Record: VALIDATED

2026-05-23T23:10:48.999916+00:00

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Proof ID: `183f038614d6a6288a95be163df4a55a…`
Created: 2026-05-23T23:10:48.999916+00:00

1. Claim

> Across a trajectory with spaced re-exposures, the substrate keeps recognising every re-exposed stimulus: (for every same-stimulus re-exposure pair (i, j) in the trajectory, Jaccard(concepts_i, concepts_j) $\geq$ 0.80). The reported floor is the worst pair across the whole run.

- **Operationalisation:** Run a schedule of len(stimuli) unique stimuli each repeated 3× and interleaved (gap = len(stimuli)) through Kimera's entity target (Takwin). For each stimulus, take the per-cycle "concepts" set at each exposure and compute Jaccard over all exposure pairs. recognition_jaccard_floor = min over all pairs across all stimuli; the LTL invariant holds iff floor $\geq \theta$.
- **Threshold:** `recognition_jaccard_floor >= 0.8 jaccard`
- **H0:** H0: floor < 0.80 — recognition collapses under manifold deformation somewhere in the run
- **H1:** H1: floor $\geq$ 0.80 — recognition is stable across the whole trajectory (memory-as-deformation holds as a flow property)

2. Verdict

VALIDATED — observed `0.95`
recognition floor 0.9500 over 24 re-exposure pairs (8 stimuli × 3 exposures); mean Jaccard 0.9958; 0 exposures produced no concept set; worst pair: stimulus 2 cycles 2↔10 = 0.9500; control: same-stimulus median 1.000 vs cross-stimulus median 0.026, Mann-Whitney p=1.7e-17 (recognition IS significantly above the cross-stimulus baseline)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
O.flow.recognition_stability	recognition_jaccard_floor	0.95	—	ophamin 0.115.2

4. Pre-registration

- Registered at: 2026-05-23T23:10:48.762737+00:00
- Config hash: 0fbe8ce56b2a2db7203910b72aaebc0c…
- Data hash: 4e6090352a40ea27df1cf08fb4d02bd5…

5. Signature

985f6a3af07e8039602a6e9ffebc654fabaff79c326d9ea18478fd009256fc29